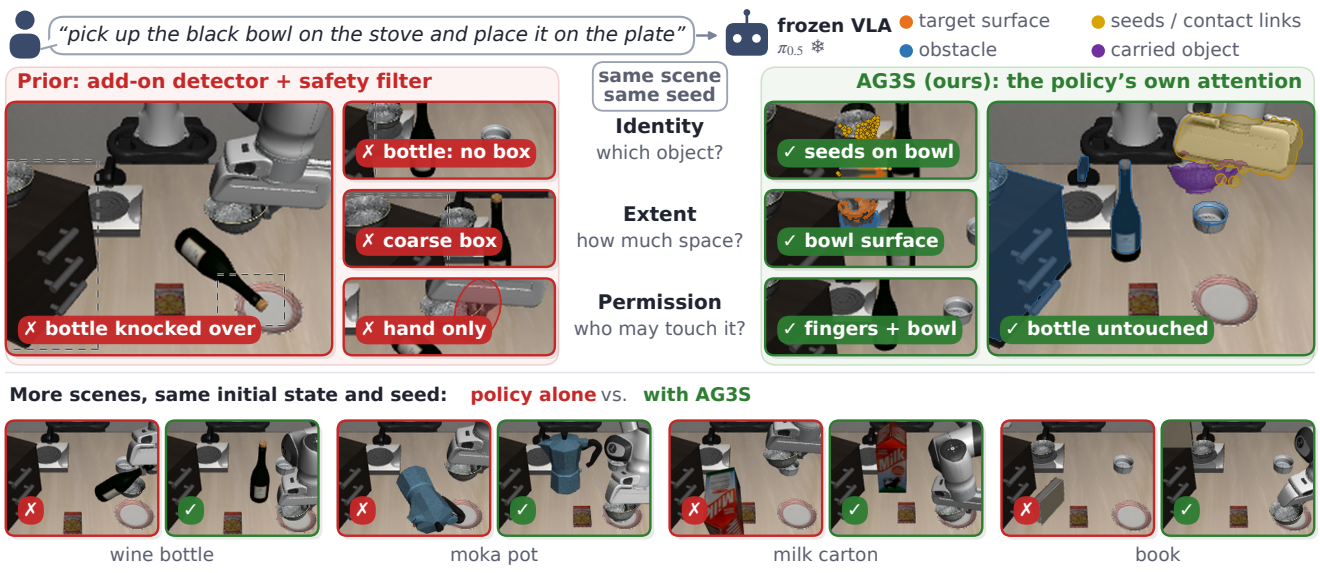

“pick up the black bowl on the stove and place it on the plate”
frozen VLA
$\pi_{0.5}$ ❄
target surface
obstacle
seeds / contact links
carried object
Prior: add-on detector + safety filter
✗ bottle knocked over
✗ bottle: no box
✗ coarse box
✗ hand only
same scene
same seed
Identity
which object?
Extent
how much space?
Permission
who may touch it?
AG3S (ours): the policy’s own attention
✓ seeds on bowl
✓ bowl surface
✓ fingers + bowl
✓ bottle untouched
More scenes, same initial state and seed: policy alone vs. with AG3S
wine bottle
moka pot
milk carton
book